1.4 Modern and smart materials, composite materials and technical textiles

1.4.1 Modern and smart materials

REVISED

Smart materials change or react to a change in their environment, such as temperature, light, pressure or electrical input. Reactions include a change in colour, shape or resistance.

Shape-memory alloys

- Shape-memory alloys (SMAs) appear to have a memory that allows them to return to an original shape if heated.
- Nitinol, an alloy of nickel and titanium, is a commonly used SMA. Possible uses include medical applications such as medical fastenings used in bone fractures.

Nanomaterials

- Nanomaterials are defined as having particles on nanoscale dimensions, approximately equivalent to one billionth of a metre.
- In textiles, nanoparticles can alter the properties of a fabric, for example adding a stain-resistant finish.
- New materials have been developed as a result of miniaturisation of component parts, which has led to the development of **conductive fibres**, commonly referred to as **e-textiles**.
- Conductive fibres and threads developed from carbon, steel and silver can be woven into textile fabrics and made into clothing, or the conductive threads can be sewn into a product to connect a circuit. This allows interaction between the fabric and the user.

Reactive glass

- Reactive glass reacts almost immediately to bright sunlight. It is controlled by a **light-dependent resistor (LDR)**.
- Smart glass reacts in a similar way but has to be activated by a switch. It is opaque when switched off. It is used in homes to give privacy instead of using curtains or blinds.

Piezoelectric materials

- Piezoelectric materials create an electrical charge when exposed to compression or pressure. Possible uses include sound or sonar detection and lighters for grills and barbeques.

Temperature-responsive polymers

- Photochromic pigments or dyes change colour in response to changes in light. For example, sunglasses can change colour in response to UV radiation.
- Thermochromic pigments or dyes change colour in response to a change in heat and can be engineered to specific heat ranges.

Conductive fibres: fibres that conduct electricity.

E-textiles: textile fabrics that have integrated and interactive devices built into the fabric that react with the user.

Light-dependent resistor (LDR): a sensor input that can detect changes in light levels.



Figure 1.6 Flexible solar panels can now be integrated into textiles fabrics



Figure 1.7 Thermochromic mugs change colour when boiling water is poured in

Conductive inks

Conductive inks contain conductive particles that allow designers to hand-draw or digitally print circuits that are fully functional.

1.4.2 Composites

REVISED

Composites are created when two or more materials are joined together to create a new, enhanced material. One material is known as the **matrix** while the other is the **reinforcement**.

Composite material: a new material created by combining two or more different types of material.

Compressive strength: the ability to withstand a pushing (squashing) force.

Concrete

- Concrete is made up of various aggregates (gravel, rock or sand), cement and water. The cement binds the materials together when water is added. Concrete has a high **compressive strength** but is much weaker under tension.
- Uses include the construction of buildings and roads, as well as smaller projects such as kerbs and drainage.

Plywood

- Plywood is made up of several layers of wood glued on top of each other. Each layer is laid at a 90-degree angle to the previous layer, giving plywood a consistent strength. The outside layers might be a more expensive wood such as birch.
- Uses include roofing, flooring and furniture.



Figure 1.8 Plywood

Fibre/carbon/glass and reinforced polymers

- Carbon fibre reinforced polymer (CFRP) is a composite consisting of woven carbon fibre strands encased in a polymer resin.
- Carbon fibre strands have a high tensile strength and the polymer resin is lightweight and rigid; when combined, this creates a high-performance engineered material. Possible uses include high-performance sports equipment.
- Glass reinforced plastic (GRP) is a composite of glass fibres and a polyester resin. Glass fibres create rigidity and the resin makes GRP tough and lightweight.
- GRP is difficult to recycle because the process of combining the glass fibres and resin cannot be reversed.

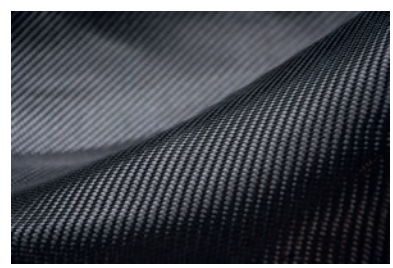


Figure 1.9 CFRP matting

Robotic materials

- Scientists are developing materials that will appear to 'think' for themselves. It is anticipated that these materials will sense something, process it, then react to that sensation, although this is some way off from being a reality.
- Possible uses could include camouflage or load balancing.

1.4.3 Technical textiles

REVISED

- Technical textiles are engineered with specific performance characteristics that suit a particular purpose or function.
- Technical textiles can be woven or bonded, synthetic or natural.

Table 1.8 Technical textiles and their uses

Textile	Where used	Uses
Agro-textiles	Agricultural industry	Mainly used in crop protection to suppress weeds or for moisture control
Construction textiles	Construction industry	Netting used on scaffolding to protect passers-by from falling debris; linings for swimming pools and ponds
Geotextiles	Civil engineering, road construction and building	Permeable fabrics used with soil that have the ability to filter, separate, protect and drain; control of embankments on the sides of roads and protection to prevent coastal erosion

Domestic textiles

- Domestic textiles have various applications in the home, from padding inside soft furnishing to cleaning cloths.
- **Microfibres** are engineered fibres that are about 100 times smaller than a human hair. When used in cleaning clothes they are highly effective in attracting dust.

Microfibres: tiny fibres about 100 times thinner than a human hair.

Environmentally friendly textiles

- Environmentally friendly textiles come from sustainable sources such as plants and animals.
- However, to be considered truly eco-friendly they have to be grown organically, without the use of pesticides and fertilisers to boost crops.

Protective textiles

- Gore-Tex is the most well-known breathable fabric and consists of three or more fabrics laminated together with a breathable **hydrophilic membrane** in the middle:
  - Warm air and tiny droplets of moisture from perspiration can permeate out through the breathable membrane but moisture from larger rain droplets and wind cannot enter.
  - It is used in high performance clothing and footwear, and it helps to regulate body temperature.
- **Aramid fibres** are engineered fibres with very high tensile strength and resistance to heat.

Hydrophilic membrane: a solid structure that stops water passing through but, at the same time, can absorb and diffuse fine water vapour molecules.

Aramid fibres: engineered synthetic polymers that are very strong and resistant to heat.

Exam tip

Be able to fully explain how smart, composite and technical textiles are used in the design and manufacture of products. Know their specific properties and what makes each suitable for a particular purpose in a product.

Typical mistake

Don't confuse technical textiles such as Gore-Tex with fabrics such as cotton that absorb moisture making them appear breathable.

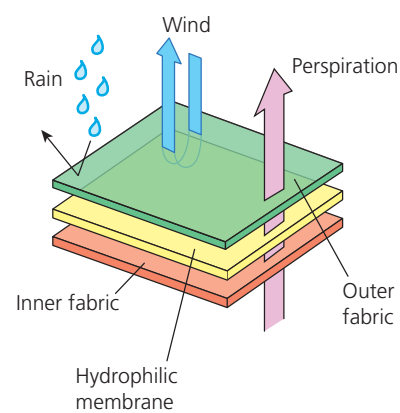


Figure 1.10 Gore-Tex is an example of a laminated fabric